Reproduction methodology — fine-tuning Whisper for Pashto ASR
CS-444 Assignment 2 · reproduction of Rahman (2026), arXiv:2604.06507

1 · Base paper and its released artefacts

Rahman (2026) — “Fine-tuning Whisper for Pashto ASR: strategies and scale”
arXiv:2604.06507 · DOI 10.48550/arXiv.2604.06507 · 7 April 2026 · preprint, cs.CL

Reproduction target — exp_009
whisper-small, full fine-tuning, Common Voice v24
Reported: WER 24.89 % · CER 8.075 %
Zero-shot baseline reported: 120.4 % WER

Released artefacts (Apache-2.0)
ihanif/exp_009_small_cv24_vanilla
experiment_log.txt · training_args.bin
Training code and corpora: NOT public (HTTP 401)

parsed

Configuration recovered by us
lr 1e-4 · true batch 64 · FP32 · warmup 500
weight decay 0.0 · patience 5 · 0.5–30 s
eval_dataset_splits = TEST (selection on test)

2 · Data preparation

Mozilla Common Voice Scripted Speech 24.0 — Pashto (ps), CC0-1.0
Official HF mirrors empty since Oct 2025 → obtained from Mozilla Data Collective / CC0 Kaggle mirror

Official shipped splits — no re-split
train.tsv / dev.tsv / test.tsv
speaker-disjoint by construction; verified

Filters (paper + recovered config)
non-empty · Arabic script · de-duplicated
≤ 448 label tokens · 0.5–30 s · 16 kHz mono

Train (subset)
~12 h seeded, seed 42
paper used 113.4 h

Dev — held OUT
1,000 utterances
paper merged dev into train

Test — untouched
13,791 utterances / 20.7 h
used once, at the end

3 · Model and the three experimental arms

train *select* *test audio*

openai/whisper-small — 244 M parameters, encoder–decoder Transformer
80-channel log-Mel, 30 s window → 12 encoder layers → 12 decoder layers → 51,865 BPE · prompt ⟨|ps|⟩ ⟨|transcribe|⟩

B0 · Zero-shot
untouched weights
no training
target: >100 % WER
paper: 120.4 %

B · Author checkpoint
his weights, our harness
no training
target: 24.89 % WER
8.075 % CER

C · Our fine-tuning
full FT · AdamW · lr 1e-5
eff. batch 32 · fp16
grad-checkpointing
best on val WER, patience 5

Why arms B and C are separate: B tests whether the paper's REPORTED NUMBER is verifiable (measurement).
C tests whether its TRAINING RECIPE reproduces at our compute budget. Different claims, different answers.

4 · Evaluation, our own analyses, and comparison

Identical decoding for all three arms — greedy, max_new_tokens 225, language = ps

Metrics
BasicTextNormalizer (paper's normaliser)
WER_norm ← headline · WER_ortho · CER
bootstrap 95 % CI · S/D/I breakdown

D · Our own analyses (not in the paper)
D1 Pashto-script output rate
D2 orthographic-normalisation ablation
paired bootstrap, zero-shot vs fine-tuned

Difference vs the paper = ours − 24.89 % (percentage points and relative) → Figures 1–8, metrics.csv, report §§6–9